

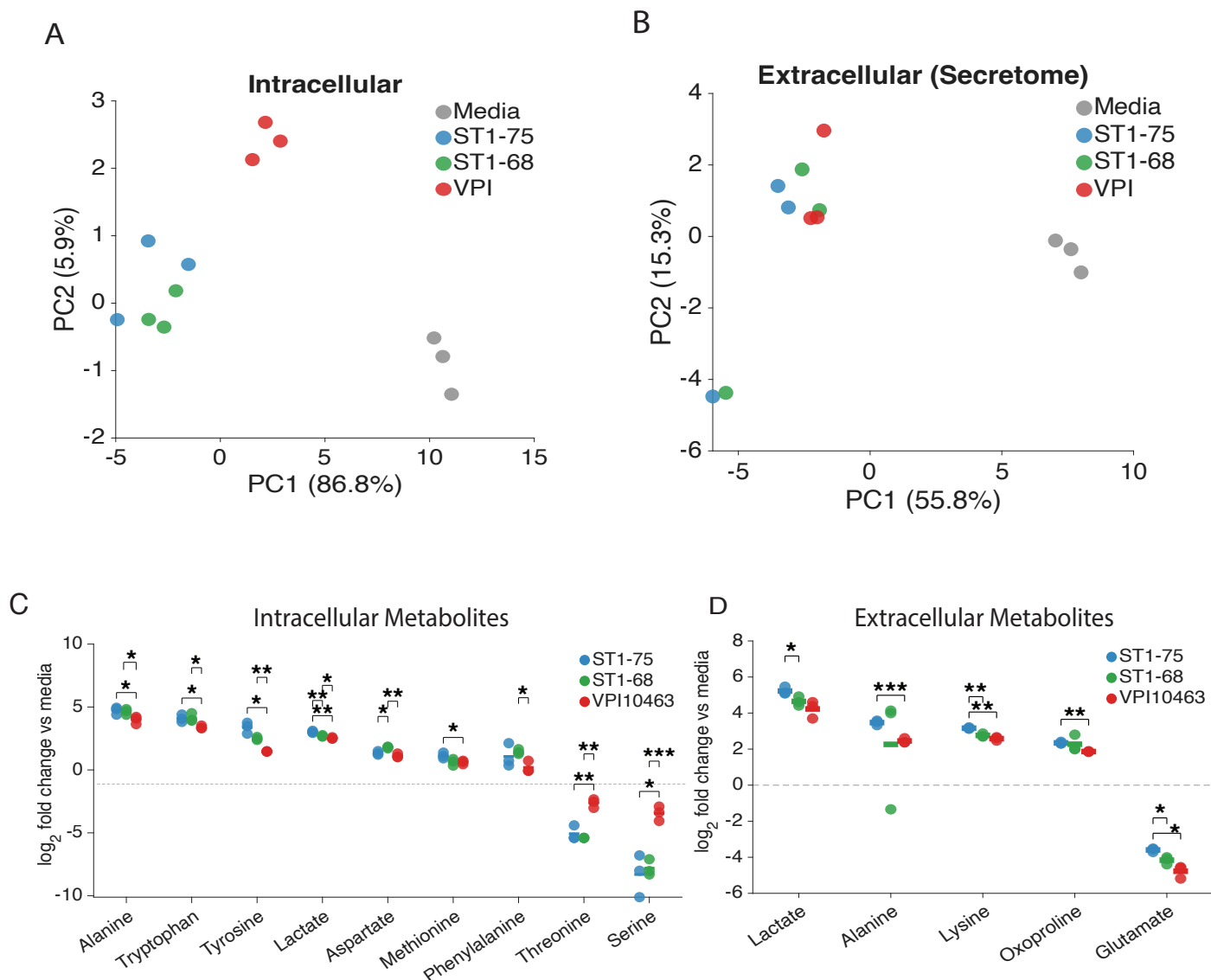


Figure 6: PCA of GC-MS metabolite profiles reveals distinct metabolic signatures across bacterial strains. (A) PCA of intracellular metabolites from the same strains. PC1 and PC2 explain 86.8% and 5.9% of the variance, respectively. Each dot represents one biological replicate. (B) Principal component analysis (PCA) of extracellular metabolites (secretome) from ST1.75 (blue), ST1.68 (green), VPI10463 (red), and media-only controls (gray). PC1 and PC2 explain 55.8% and 15.3% of the total variance, respectively. (C) Intracellular metabolite log₂ fold change relative to plain CDMM media for ST1.75 (blue), ST1.68 (green), and VPI10463 (orange), assessed by GC-MS following TMS derivatization of cell pellets from 6-hour cultures at comparable biomass. Only metabolites showing a statistically significant difference in at least one pairwise comparison are shown (Welch's t-test, * p < 0.05, ** p < 0.01, *** p < 0.001). Metabolites are ordered from most accumulated (left) to most depleted (right) relative to media. Error bars indicate SD (n = 3). (D) Extracellular metabolite log₂ fold change relative to plain CDMM media for the same strains and conditions as in (C), assessed from spent media. Error bars indicate SD (n = 3).